

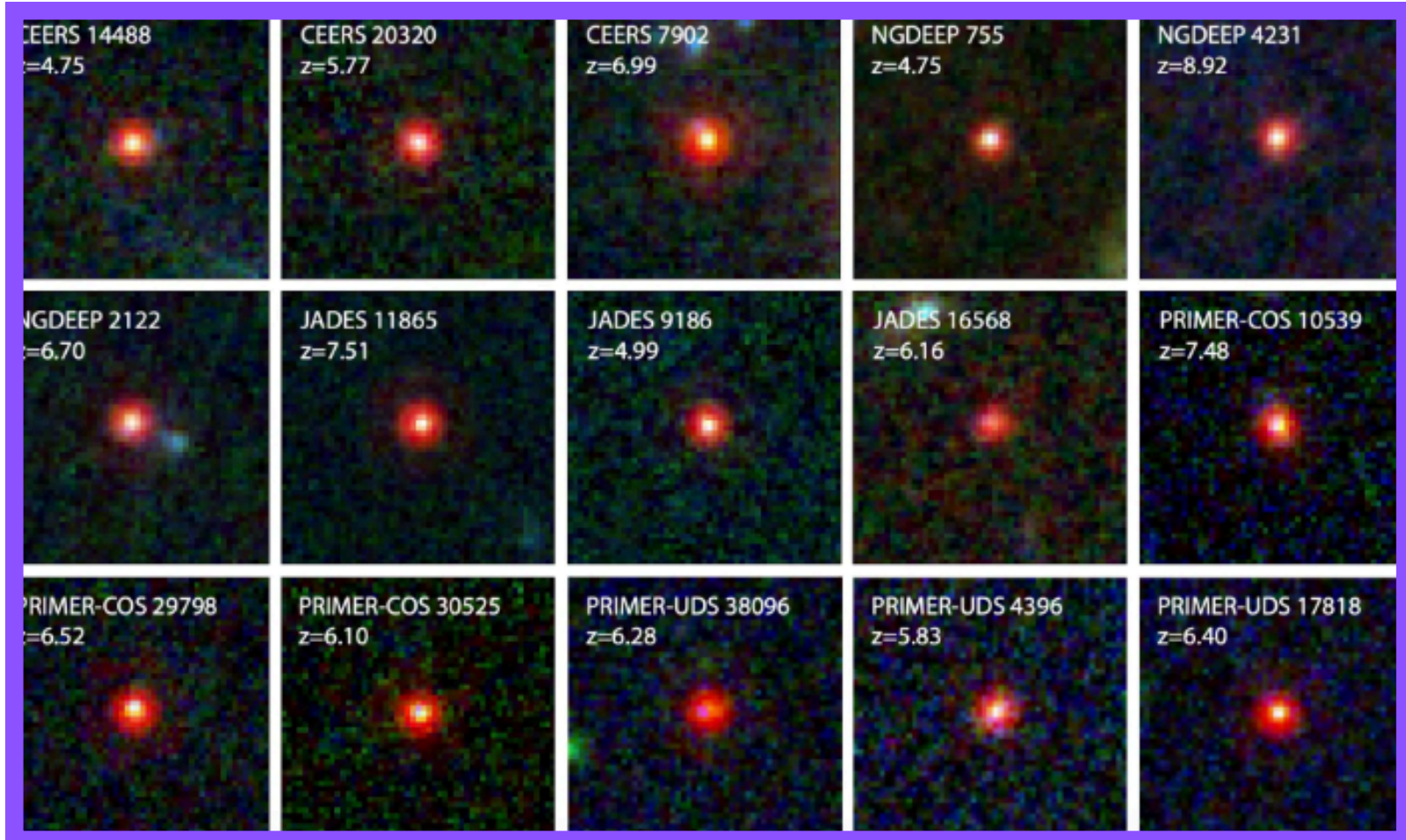
# A Comparative Study of Primordial and Stellar Seed Growth in JWST Little Red Dots



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## Introduction:

- JWST discovered early massive black holes in Little Red Dots (LRDs)
- Small stellar-remnant seeds may not grow fast enough to explain them
- Larger primordial black hole seeds could better match the observations



Kocevski et al., 2025

Examples of JWST Little Red Dots (LRDs) across  $z \sim 4-7$ , showing compact, red sources

Access this data:

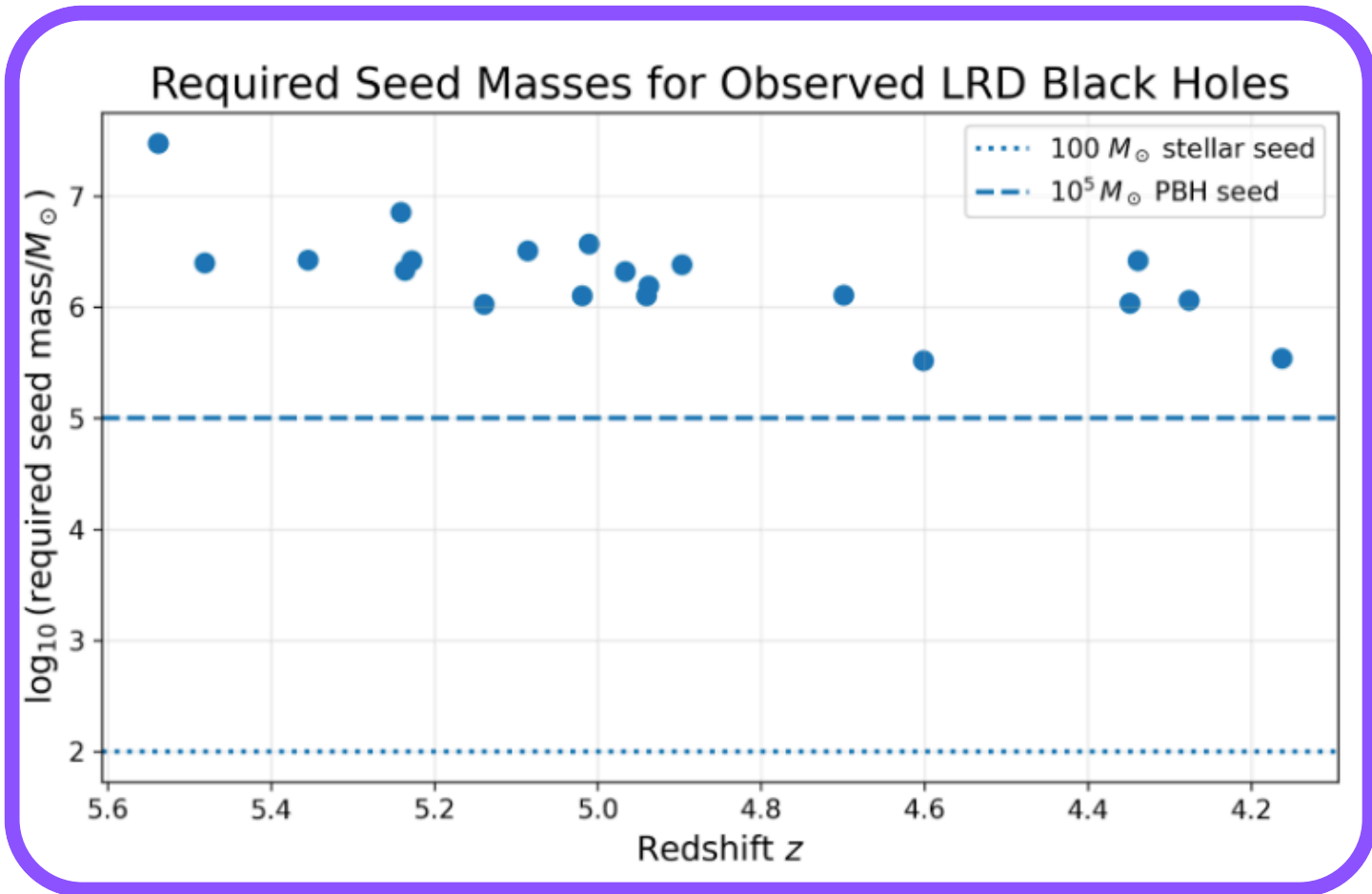


## Data and Results:

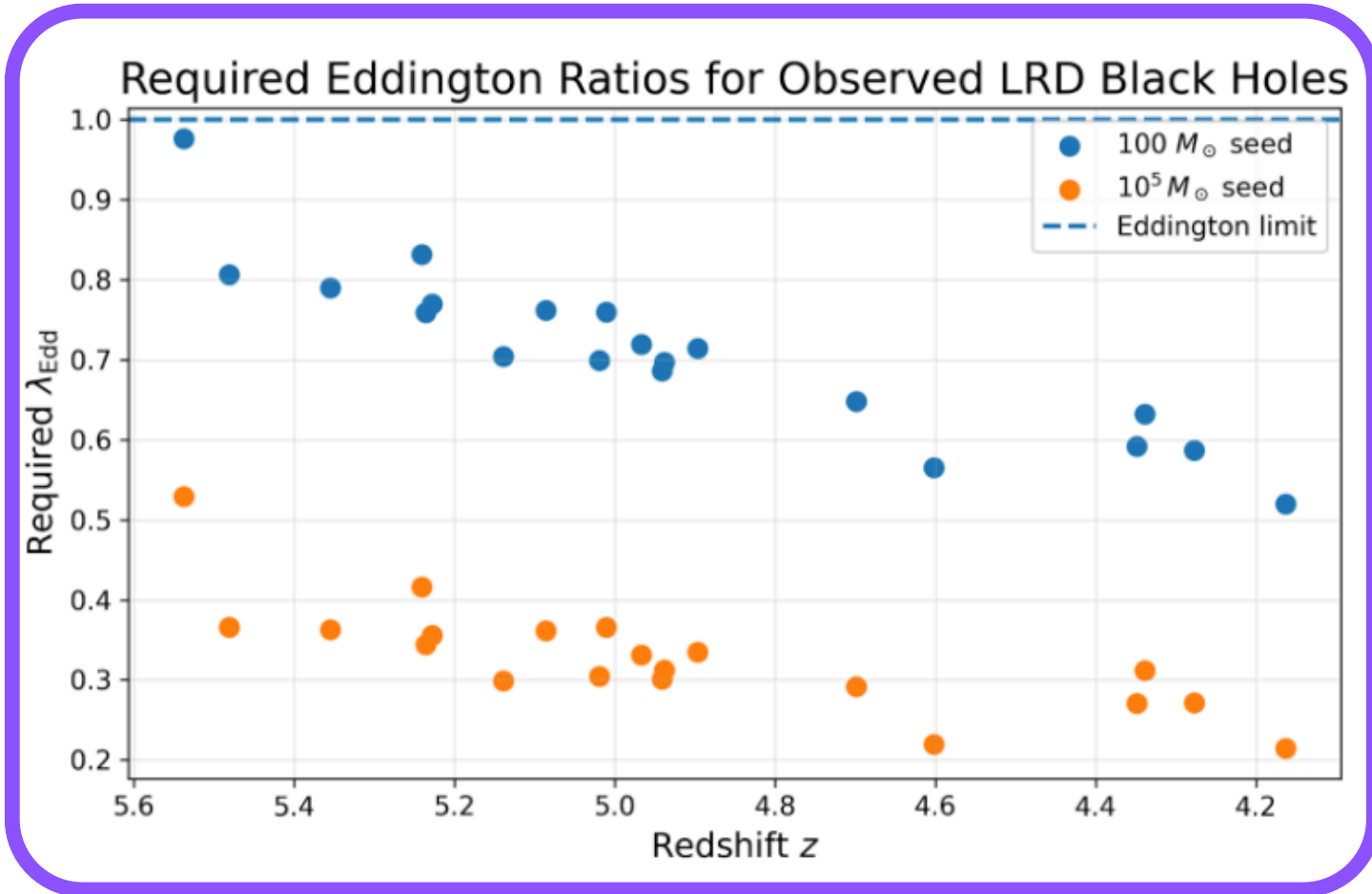
- Stellar-remnant seeds usually do not grow large enough to match the black holes observed in LRDs
- Observed masses require much larger initial seeds than standard stellar models provide
- Primordial black hole seed models more successfully reproduce the observed black hole populations without unrealistic growth rates

## Methods:

- Black hole growth simulated from early galaxies to later cosmic times using A-SLOTH semi-analytic model
- Includes mergers, gas accretion, feedback effects, and burst growth events
- Growth compared for both stellar-remnant and primordial black hole seed scenarios under Eddington-limited accretion



The seed mass required to reproduce an observed LRD black hole under realistic growth assumptions



Eddington ratios needed for different seed masses to grow into observed LRD black holes, indicating accretion efficiency

Access this data:



| ID   | Field      | R.A.<br>[deg] | Dec<br>[deg] | {filt}_flux<br>[μJy] | {filt}_fluxerr<br>[μJy] | z_phot                                 | A_V<br>[mag] | log(L_bol/erg/s) | M_UV<br>[ABmag] | f_Edd_pix<br>[pc] | f_Edd_galpy<br>[pc] |
|------|------------|---------------|--------------|----------------------|-------------------------|----------------------------------------|--------------|------------------|-----------------|-------------------|---------------------|
| 1381 | PRIMER-UDS | 34.440988     | -5.209337    | -                    | -                       | 5.66 <sup>+0.02</sup> <sub>-0.02</sub> | 3.1 ± 0.3    | 45.0 ± 0.1       | -17.52 ± 0.3    | < 1.04            | < 186               |
| 1470 | PRIMER-UDS | 34.260549     | -5.209163    | -                    | -                       | 5.40 <sup>+0.06</sup> <sub>-0.06</sub> | 1.1 ± 0.1    | 44.4 ± 0.2       | -18.45 ± 0.1    | < 1.23            | < 230               |
| 1544 | PRIMER-UDS | 34.377413     | -5.209023    | -                    | -                       | 5.46 <sup>+0.02</sup> <sub>-0.02</sub> | 1.0 ± 0.2    | 44.5 ± 0.4       | -18.84 ± 0.1    | < 1.02            | < 187               |
| 1939 | PRIMER-UDS | 34.403098     | -5.208186    | -                    | -                       | 4.36 <sup>+0.12</sup> <sub>-0.12</sub> | 1.3 ± 0.1    | 44.6 ± 0.4       | -17.78 ± 0.1    | < 0.71            | < 144               |
| 3992 | PRIMER-UDS | 34.367304     | -5.204196    | -                    | -                       | 5.66 <sup>+0.02</sup> <sub>-0.02</sub> | 1.6 ± 0.2    | 45.2 ± 0.2       | -18.53 ± 0.1    | < 0.55            | < 104               |
| 7991 | PRIMER-UDS | 34.293687     | -5.196477    | -                    | -                       | 5.66 <sup>+1.38</sup> <sub>-0.19</sub> | 2.4 ± 0.3    | 45.9 ± 0.4       | -18.47 ± 0.3    | < 1.11            | < 196               |
| 8559 | PRIMER-UDS | 34.465426     | -5.195333    | -                    | -                       | 7.94 <sup>+0.09</sup> <sub>-0.09</sub> | 3.9 ± 0.1    | 46.7 ± 0.2       | -19.50 ± 0.1    | < 0.68            | < 111               |
| 0404 | PRIMER-UDS | 34.462872     | -5.191960    | -                    | -                       | 5.66 <sup>+0.02</sup> <sub>-0.02</sub> | 2.2 ± 0.2    | 45.6 ± 0.3       | -18.13 ± 0.1    | < 0.50            | < 101               |
| 1219 | PRIMER-UDS | 34.471126     | -5.190444    | -                    | -                       | 5.43 <sup>+0.02</sup> <sub>-0.02</sub> | 2.7 ± 0.2    | 45.9 ± 0.3       | -19.20 ± 0.1    | < 0.42            | < 84                |
| 4419 | PRIMER-UDS | 34.399178     | -5.184323    | -                    | -                       | 4.82 <sup>+0.27</sup> <sub>-0.27</sub> | 0.9 ± 0.2    | 44.5 ± 0.3       | -17.84 ± 0.1    | < 0.64            | < 135               |
| 5257 | PRIMER-UDS | 34.431864     | -5.182780    | -                    | -                       | 6.92 <sup>+0.06</sup> <sub>-0.06</sub> | 1.3 ± 0.2    | 45.2 ± 0.1       | -18.85 ± 0.2    | < 0.79            | < 133               |
| 6999 | PRIMER-UDS | 34.508058     | -5.180084    | -                    | -                       | 4.21 <sup>+0.12</sup> <sub>-0.12</sub> | 1.1 ± 0.1    | 45.2 ± 0.2       | -18.41 ± 0.1    | < 0.65            | < 154               |
| 8804 | PRIMER-UDS | 34.363515     | -5.176964    | -                    | -                       | 4.32 <sup>+0.07</sup> <sub>-0.07</sub> | 1.5 ± 0.1    | 45.1 ± 0.4       | -19.21 ± 0.1    | < 0.45            | < 114               |
| 8892 | PRIMER-UDS | 34.268908     | -5.176722    | -                    | -                       | 4.73 <sup>+0.03</sup> <sub>-0.03</sub> | 1.0 ± 0.1    | 44.4 ± 0.2       | -17.34 ± 0.1    | < 0.62            | < 123               |
| 9416 | PRIMER-UDS | 34.460763     | -5.175813    | -                    | -                       | 5.43 <sup>+0.02</sup> <sub>-0.02</sub> | 1.9 ± 0.1    | 44.8 ± 0.3       | -18.32 ± 0.1    | < 1.07            | < 221               |
| 1108 | PRIMER-UDS | 34.362963     | -5.173290    | -                    | -                       | 7.14 <sup>+1.8</sup> <sub>-0.16</sub>  | 2.6 ± 0.3    | 45.8 ± 0.2       | -18.09 ± 0.3    | < 0.62            | < 105               |
| 2773 | PRIMER-UDS | 34.438986     | -5.170543    | -                    | -                       | 4.40 <sup>+0.23</sup> <sub>-0.10</sub> | 1.3 ± 0.2    | 44.7 ± 0.2       | -18.88 ± 0.1    | < 0.57            | < 129               |
| 3575 | PRIMER-UDS | 34.408020     | -5.169147    | -                    | -                       | 5.66 <sup>+0.02</sup> <sub>-0.02</sub> | 2.0 ± 0.2    | 45.1 ± 0.3       | -18.20 ± 0.1    | < 0.76            | < 137               |
| 4081 | PRIMER-UDS | 34.346207     | -5.168197    | -                    | -                       | 5.66 <sup>+0.02</sup> <sub>-0.02</sub> | 1.8 ± 0.2    | 45.1 ± 0.3       | -18.42 ± 0.1    | < 0.52            | < 100               |

Kokorev et al., 2024

LRD data used to constrain stellar- and primordial black hole seed growth

## Discussion and Conclusions:

- Stellar-remnant seeds struggle to reach the black hole masses observed in JWST Little Red Dots
- Many LRDs require much larger initial seed masses
- Stellar seeds would need unrealistic near-Eddington growth rates to match observations
- Heavy-seed models, such as primordial black holes, more naturally explain the observed populations